Rhea Rajmanna

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Education

Virginia Polytechnic Institute and State University — Blacksburg, VA
General Engineering w/ Interest in Computer Science — *Aug. 2024 – Present (Expected May 2027)*

Chantilly High School — Chantilly, VA
Advanced Diploma; GPA (Weighted): 4.46/4.00 — *Aug. 2020 – June 2024*

Relevant Coursework

Current: Software Design and Data Structures, Discrete Mathematics, Multivariable Calculus, Introduction to Linear Algebra, Foundations of Engineering, Foundations of Physics, Technical Writing, **Past**: AP BC Calculus (5), AP AB Calculus (5), AP Research (5), AP Computer Science A (4), AP Computer Science Principles (5)

Technical Skills

Languages: Java, Python, SQL, JavaScript, HTML/CSS, **Developer Tools**: VS Code, PyCharm, IntelliJ, Eclipse, **Libraries**: Pandas, NumPy, Matplotlib

Work/Research Experience

Tim Duong Research Laboratory, Albert Einstein College of Medicine — June 2024 – Present
Student Researcher — *Remote, Bronx, NY*

- Querying Montefiore patient database with SQL to analyze COVID-era blood pressure trends.
- Developing data visualizations to track BP shifts across age, gender, and race groups.
- Collaborating with medical students to interpret statistical findings.

Virginia Tech Transportation Institute — March 2025 - Present
Undergraduate Research Assistant — *Blacksburg, VA*

- Project: Assessment of Drowsiness Using the Johns Drowsiness Scale (JDS): An Objective, Continuous Index Using Eyelid Movement Measures
- Analyzing 75+ hours of driver video to assess drowsiness using JDS and PERCLOS metrics.
- Statistical analysis to correlate JDS with physiological and behavioral drowsiness indices (e.g., KSS, lane deviations, heart rate).

Aspiring Scientists Summer Internship Program (ASSIP) — June 2023 – Jan. 2024
George Mason University, Dr. Wijesekera, Dept. of Computer Science and Cybersecurity — *Arlington, VA*

- Built Python data fetching algorithm and animated motion trail using Grid Search machine learning algorithm.
- Researched implementation of sensor-based occupancy detection in the CLI Innovation Lab to produce an energy-efficient HVAC system.
- Wrote an abstract "Occupancy and Direction Estimation Based on Grid Search Approach."

Projects

NutriCampus | *React, Postman, HTML, CSS* | nutricampus.vercel.app — Sept. 2024

- Built NutriCampus, a React.js platform offering personalized meal recommendations from campus dining menus.
- Integrated with Michigan Dining's API to organize and access data.
- Designed UI/UX using HTML and CSS.

Optimism's Effect on Ethereum Model | *Blockchain, Statistical Data Analysis* — July 2023

- Analyzed optimism's effect on ethereum, collected transaction data from Ethereum and Optimization Explorers.
- Published abstract "Assessing Optimism's Impact on Ethereum's Scalability and Transaction Efficiency"
- Received the 2023 Conrad Innovator Award.

Awards and Honors

AP Capstone Diploma, National Merit Finalist, AP Scholar With Distinction